Module 4: Cells - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. State the core claims of cell theory.
2. Compare prokaryotic and eukaryotic cells using structure and scale.
3. Explain how membranes and compartments support cell function.
4. Match major organelles to their roles.
5. Use microscope evidence to support claims about cells.

Topics

- Cell theory
- Prokaryotic and eukaryotic organization
- Membranes and compartments
- Organelles and division of labor
- Microscopy and evidence

Key Terms to Know

- **Cell theory** - Principle that organisms are made of cells and cells come from cells.
- **Prokaryote** - Cell without a membrane-bound nucleus.
- **Eukaryote** - Cell with a membrane-bound nucleus and organelles.
- **Organelle** - Specialized cell structure with a particular job.
- **Nucleus** - Eukaryotic compartment that stores DNA.
- **Mitochondrion** - Organelle central to aerobic energy harvesting.
- **Chloroplast** - Plant and algal organelle that performs photosynthesis.
- **Microscopy** - Use of lenses or imaging tools to observe small structures.

Core Contents

1. Cell theory explains cells as the basic units of life and as descendants of existing cells.
2. Prokaryotic and eukaryotic cells differ in compartmentalization and internal organization.
3. Membranes separate internal conditions from the environment and define organelles.
4. Organelles divide cellular work such as energy capture, protein processing, and storage.
5. Microscopy turns cell structure into observable evidence.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-04_liquid-chemistry.md`

Generated Visuals

- Module 04: Cells Evidence Map
- Module 04: Cells Reasoning Sequence
- Module 04: Cells Retrieval and Lab Check